

MPV MEDIA PLAYER — COMPREHENSIVE USER GUIDE

1. Introduction & System Overview

biraj-mpv-conf is a clean, optimized configuration suite designed for mpv. It brings a modern user interface, smooth 4K video playback, fast hover thumbnails, and native file dialogs into a single seamless experience.

The key features include:

- **High-Performance Rendering:** Uses the `gpu-next` engine with Direct3D 11 Flip presentation for low GPU overhead.
- **HDR10 & Dolby Vision Tone-Mapping:** Real-time tone mapping tuned for 0-frame-drop playback on standard laptop screens.
- **Dynamic Format Badge Overlay:** Automatic on-screen notification of detected video formats (`DV` , `HDR10+` , `HDR` , `HLG` , `SDR`).
- **Modern UI:** ModernZ On-Screen Controller with fluent vector icons and interactive right-click context menu.
- **Smart Contextual Profiles:** Automatic profile switching for Picture-in-Picture mode, minimizing, and night audio normalization.

2. Installation & Setup

Installation Mode	Target Directory Path
Standard (Roaming)	<code>C:\Users\<Username>\AppData\Roaming\mpv\</code>
Portable Mode	<code>C:\<PathToMpv>\portable_config\</code>

3. Video Playback & HDR Performance Tuning

- **Tone-Mapping Engine:** Configured with `vo=gpu-next` and `d3d11-flip=yes` for low-overhead DirectX flip-model presentation.
- **Compute Peak Optimization:** Setting `hdr-compute-peak=no` uses stream metadata directly, reducing render time to ~ 25 ms and preventing dropped frames during 4K Remux playback.
- **Display Hinting:** Enables `target-colorspace-hint=yes` to pass wide-gamut BT.2020 metadata directly to external HDR displays.

4. Dynamic Format Badge Overlay

The included `hdr_badge.lua` script automatically detects incoming video properties and announces the format in the top-right corner:

- **DV:** Dolby Vision Profiles 5 and 8.1 streams
- **HDR10+:** Dynamic metadata HDR streams
- **HDR:** Standard PQ and BT.2020 high dynamic range content
- **HLG:** Hybrid Log-Gamma broadcast streams
- **SDR:** Standard Dynamic Range content

The badge displays for 5 seconds on launch and can be manually viewed or hidden at any time by pressing `1`. When the player is muted, it automatically shifts left to sit beside the speaker icon without collision.

5. Smart Automation Profiles

Profile Name	Trigger Condition & Behavior
<code>[Window-PiP]</code>	Windowed on-top mode: strips window borders and scales OSC by 1.8x
<code>[Minimized]</code>	Auto-pauses playback when the player window is minimized
<code>[Video]</code>	Enables live video playback progress on the Windows taskbar
<code>[Image]</code>	Sets infinite display duration and enables cursor-centered zoom
<code>[night-audio]</code>	Dynamic audio normalizer balancing dialogues and sound effects
<code>[dolby-vision-perf]</code>	Low-overhead tone-mapping profile for smooth 4K DV/HDR playback